

Eye Ball: Gazing as a Dilemma in a Competitive Virtual Reality Game

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Figure 1: The competitive VR game “Eye Ball” introduces a novel gaze-based mechanic, creating strategic dilemmas in gameplay.

ABSTRACT

In the evolving landscape of gaming technology, our research explores the integration of eye-tracking in VR games, moving beyond traditional applications. This paper presents “Eye Ball,” a competitive VR game that introduces a novel gaze-based mechanic, creating strategic dilemmas in gameplay. Our approach utilizes the qualities of gaze behavior to enhance social interaction in games, offering a more immersive gaming experience. In “Eye Ball,” players can generate a shield by closing their eyes, providing protection at the strategic cost of visual awareness. This mechanic challenges traditional game design, balancing the power of protection with the vulnerability of reduced situational awareness. A comparative study was conducted to examine the effects of this gaze-based dilemma. Our findings demonstrate the potential of gaze mechanics in enhancing the depth of social VR games. Our research contributes to the

broader field of HCI, highlighting the potential of gaze interaction in creating immersive and socially complex gaming experiences.

CCS CONCEPTS

• Human-centered computing;

KEYWORDS

Gaze-based interaction, VR games, social games

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1 INTRODUCTION

Recent advancements in eye-tracking technologies have significantly influenced the gaming landscape, offering novel ways to enhance player interaction and experience. This paper explores the evolving role of eye-tracking in games, not just as a tool for analyzing user attention but as a transformative element in game design and interaction. While the integration of eye-tracking in various

technical devices, such as computer monitors and VR/MR headsets, has been explored for enhancing technical performance (e.g., foveated rendering) and immersive experiences (e.g., avatar interactions), its potential in gaming contexts remains largely untapped [25–27].

Despite the proliferation of eye-tracking-supported games, there is a noticeable absence of a 'killer app' or an innovative game concept that could catalyze a breakthrough for this technology in the gaming sector [33]. Our work addresses a specific aspect of this potential: leveraging eye-tracking as a game mechanic that respects and utilizes the natural qualities of the human gaze. Traditionally, eye-tracking has been employed as an additive feature to established interaction methods—such as mouse, keyboard, or gamepad—often giving an impression of being a supplementary gadget rather than a core gameplay element [32].

Furthermore, the use of gaze in game design frequently manifests as a form of 'superpower,' allowing players to manipulate or destroy objects merely by looking at them [34]. This approach initially offers a novel and engaging experience but tends to lose its appeal over time, leading players to question the technology's relevance and utility [7]. We posit that gaze interaction harbors more intriguing possibilities, particularly in social settings, where gaze plays a crucial role in communication and relationship dynamics. Previous studies have touched upon this aspect but have not delved into more complex mechanics, often resorting to simplistic 'gaze-as-pointer' or 'always-on' approaches [17, 18].

In our research, we employ gaze in social gaming contexts to create strategic dilemmas. Here, the use of gaze is not always advantageous but must be strategically deployed, raising questions about what game scenarios emerge when players must selectively rely on their visual information channel, controlled through their gaze behavior. This approach is particularly compelling in competitive games, where understanding the environment and opponents is crucial [16].

The social dynamics of gaze in gaming offer a rich field for exploration. Gaze can convey a multitude of non-verbal cues, such as intent, focus, and even deception [22, 30, 36, 37]. By integrating gaze as a mechanic that players must tactically manage, we introduce a layer of social gameplay that goes beyond traditional interaction methods. This not only enhances the depth of the gaming experience but also mirrors real-life social interactions, where gaze plays a pivotal role in how we communicate and relate to others [15].

In this paper, we report on preliminary findings from a study examining the use of gaze as a dilemma mechanic in competitive VR games. We developed a comparative setting with a game prototype to investigate these dynamics. Through this study, we aim to identify potential improvements to our approach. This will enable us to conduct a more extensive study in the next phase, incorporating various iterations of the gaze dilemma and involving diverse group sizes. Our paper makes contributions to using gaze interactions in competitive games. It proposes approaches for integrating eye-tracking in game design, particularly in the context competitive play. Additionally, our research opens discussions for future exploration in this domain, providing a foundation for other researchers and game developers to build upon. The contributions of this work extend beyond game design, offering insights into the broader field

of human-computer interaction. By exploring the use of gaze in a social and competitive context, we aim at contributing to the understanding of how advanced technologies can be used to enhance not only the technical aspects of gaming but also the social dimensions. This research has the potential to influence future developments in eye-tracking technology, game design, and interactive media.

2 STATE OF THE ART

Eye-tracking technology has increasingly been incorporated into video games, spanning various applications [8, 10, 19, 23, 38]. The growing accessibility of consumer-grade eye trackers has heightened interest in gaze-based interactions within the gaming industry. These interactions have been implemented as supplementary controls alongside traditional inputs like gamepads [6], voice commands [39], hand gestures [5], or even as standalone controls for hands-free, eyes-only interaction [12], enhancing accessibility [4]. The gaming industry, recognizing the potential of eye-based interactions, has seen major players like Tobii, a leading eye-tracker device manufacturer, collaborate with game developers to integrate this technology into over 160 games, including renowned franchises and indie titles.

This surge in eye-tracking applications in games gave rise to the *EyePlay* framework [33]. Velloso and Carter [33] compiled a comprehensive list of gaze-enabled games, showcasing various interactions like object selection through gaze [6], initiating actions with eye movements [9, 13], and aiming weapons using eye direction [3]. Other games utilize gaze as a natural pointer, translating eye movements into digital interactions (e.g., rolling eyeballs in 3D and VR environments [29]), and leveraging gaze visualization to communicate or deceive other players [20].

Some researchers have focused on developing more challenging gaze-based game interactions. These designs mirror real-world eye behavior, such as creating attention dilemmas [35], moving interaction to the peripheral vision [25], or leveraging social gaze dynamics [15]. In games like *The Royal Corgi* [36], players' gaze points carry significant social implications, such as causing offense by not maintaining eye contact during conversations.

In this regard, the role of gaze in social interactions was investigated in several research projects [1, 2, 11, 14].

Shahid et al. [31], for instance, emphasized mutual gaze in enhancing social presence and understanding in both video-mediated and in-person gameplay. Newn et al. [21] showed that players can infer opponents' intentions through various gaze visualizations, which is understandable to audiences regardless of visualization type. Lankes et al. [2016] studied shared gaze visualization's impact on social presence in competitive online games, investigating its influence across genres. In cooperative tabletop gaming, Newn et al. [22] explored how projecting a player's gaze onto the board might affect strategies and tactics.

Additionally, some game designs are based on the concept of 'not looking' [25], such as making characters vanish or teleport by closing the eyes [9], or avoiding gaze contact with certain game elements to complete tasks [24]. These designs emphasize the unique control and interaction possibilities of gaze, integrating it into game narratives and enhancing player experience in terms of interaction effectiveness, immersion, and performance [28].

The current state of the art demonstrates that gaze-based interactions in games are used for both explicit control mechanisms and subtler, social, and communicative aspects. Our work builds on these insights, exploring the use of gaze as a dilemma mechanic in competitive VR games. This approach is unique in that it conceptualizes gaze not just as a pointer or an input method but as a strategic resource that must be judiciously managed. This perspective opens new dimensions in player interaction and game design, addressing a gap in the current state of the art where gaze is often underutilized or oversimplified in its application.

By integrating gaze as a mechanic that players must tactically manage, we introduce a layer of social gameplay that goes beyond traditional interaction methods. This not only enhances the depth of the gaming experience but also mirrors real-life social interactions, where gaze plays a pivotal role in communication and dynamics. Our research contributes to the understanding of gaze interaction in games, proposing new frameworks for integrating eye-tracking in game design, especially in the context of social interaction and competitive play.

3 GAME PROTOTYPE

Our game prototype, "Eye Ball", is a competitive VR game designed primarily for two players, with the potential for adaptation to bigger group scenarios. The game's design is intentionally straightforward, drawing inspiration from well-known games such as Dodgeball ¹, popular in America, and *Völkerball* ², common in some European countries. This familiarity makes "Eye Ball" accessible to a wide range of players, many of whom have encountered similar games in their childhood. In "Eye Ball," the objective is to hit the opposing player with a ball using a hand throw to score points. The game is time-limited, with each match lasting two minutes. The player with the most points at the end of this period wins. The playing field is divided into two distinct zones, with each player confined to their respective area. This division encourages strategic movement and dodging to avoid the opponent's attacks. Invisible walls surround the play area to keep the balls within bounds, ensuring a dynamic and continuous gameplay experience. Each player starts with three balls, allowing for a fluid transition between offensive and defensive play. The choice of this particular game design is twofold. Firstly, its simplicity ensures that the rules are easily understood and quickly learned, possibly through brief tutorials. Despite its straightforward rules, the game allows for complex and engaging player interactions. Players can feint attacks or dodges, and their positioning and movement within the play area can lead to intriguing gameplay scenarios.

In terms of visual representation, "Eye Ball" adopts a stylized, simplistic aesthetic, inspired by Meta's avatar design. This design choice ensures that facial expressions and gaze-based behaviors are easily recognizable on the avatars. The physics of the balls have been adjusted to enhance the 'juiciness' of controls, making the gameplay more engaging and responsive. A unique aspect of "Eye Ball" is the integration of a gaze-based dilemma into the gameplay. Players can generate a shield in front of their avatar at any point during the game, offering protection against the opponent's attacks



Figure 2: Screenshots showing the Virtual Reality game “Eye Ball”: like in Dodgeball, players can shoot balls at each other to score points (top image); players can also generate a blue shield to protect them by closing their eyes (below image).

(analogy: shield their eyes and themselves from an attack). However, this advantage comes with a strategic cost: activating the shield requires the player to close their eyes, losing visual awareness of the game and relying solely on auditory cues. The shield only offers frontal protection, allowing opponents to score by bouncing the ball off the invisible walls or timing their attacks for when the defender reopens their eyes. This mechanic introduces a nuanced layer of strategy, where players must balance the power of protection with the vulnerability of reduced situational awareness. "Eye Ball" represents an innovative approach to VR gaming, leveraging familiar gameplay mechanics while introducing a novel, gaze-based interaction that challenges traditional game design.

This prototype not only enhances the player experience through its unique mechanics but also aligns with our research goals of exploring gaze interaction as a strategic element in competitive VR games. "Eye Ball" thus contributes to the broader discourse

¹<https://www.usadodgeball.com/rules>

²<https://www.sportunterricht.ch/lektion/spielen/spielen6.php>

in human-computer interaction, particularly in the realm of integrating advanced technologies like eye-tracking into interactive media.

3.1 Technical realization

In the development of our game prototype, we employed *Unity3D 2021.3.30*³ as the primary platform for implementation. This choice was driven by Unity's robust capabilities and widespread use in the gaming industry, particularly for VR applications. Our setup included two advanced VR devices, the *Meta Quest Pro*⁴, known for their ability to detect eye movements and facial expressions. The VR experience was designed to be untethered, providing players with the freedom to move and interact within a 3 x 3-meter space. Each player was situated in the same room, which facilitated direct interaction and communication. The use of VR controllers allowed for intuitive and natural interactions with the game world. To accurately capture and interpret the players' facial features, we integrated the facial feature detection capabilities (blink detection) of the *Meta Avatars SDK 24.0*⁵. As soon as a player's eyes remained closed for longer than 0.5 seconds, an event was triggered to activate the shield. This mechanism ensured that brief intervals, potentially caused by momentary blinking, were disregarded. This design choice was crucial in maintaining the integrity of the game's mechanics, ensuring that only deliberate and sustained eye closures would initiate the shield activation. By setting this half-second threshold, we effectively differentiated between involuntary blinks and intentional actions, thereby enhancing the gameplay experience and ensuring a more accurate response to the player's physical cues. Network synchronization was achieved using *Photon (PUN 2)*⁶. This choice was made due to Photon's reliability and ease of integration with Unity, ensuring that player actions and interactions were seamlessly synchronized across the network. Further enhancing the interaction with the game world, we utilized *Oculus Integration 57.0.1*⁷. This integration provided a comprehensive suite of tools and features designed to optimize the VR experience, including improved performance, visual quality, and user interaction capabilities. The game also featured music and sound effects to enrich the gaming atmosphere. However, the volume levels were carefully chosen to ensure that players could comfortably communicate with each other during gameplay. This design decision was crucial in maintaining the social aspect of the game, allowing players to strategize, collaborate, and share their experiences in real-time.

4 COMPARATIVE STUDY

To investigate the effect of the gaze dilemma in gameplay, we conducted a comparative study consisting of two conditions. One variant allowed players to generate a shield by closing their eyes, offering frontal protection from enemy attacks. The other variant did not include this mechanic. Below, we describe the procedure, participants, and initial insights drawn from the study.

³<https://www.unity.com>

⁴<https://www.meta.com/at/en/quest/quest-pro/>

⁵<https://developer.oculus.com/downloads/package/meta-avatars-sdk/>

⁶<https://www.photonengine.com/pun>

⁷<https://assetstore.unity.com/packages/tools/integration/oculus-integration-deprecated-82022>

4.1 Procedure and Measures

We conducted a pilot study where pairs of participants could evaluate "Eye Ball". We recruited 10 participants with an age ranging from 18 to 24 years (M: 20.7; SD: 2.55) and consisting of five women and five men, who tested "Eye Ball" in 5 trials. Subjects are students from the University of California Santa Cruz and the University. The evaluation was carried in a usability lab room at the involved institution. The approach for the exploratory investigation was as follows: the experimenter welcomed the participants and explained the scope, the studies' structure, and the head-mounted devices (HMD) and its controls. After this, the participants signed a consent form and completed a survey to provide demographic data. Following, pairs of participants would playtest a version of "Eye Ball" without the gaze activated shield. Then, the pair would playtest a version of "Eye Ball" with the novel mechanic. After the two playtests, users would conclude the trial by answering the following questions on a Qualtrics form on a provided device:

- How would you characterize the difference between the two versions of the game?
- Do you prefer one version more than the other? If so, why?
- What are your impressions of the gaze mechanic (closing your eyes to create a shield)?
- Can you think of any other ways you might want to use this gaze mechanic in VR games or other applications?

In total, the experiment's duration was approximately 45 minutes per test session. This study procedure was approved by the Institutional Review Board at the University of California Santa Cruz, under protocol HS-FY2024-44.



Figure 3: Subjects playing the game: Each of the players wore an Meta Quest Pro with eye-tracking capabilities and used the controllers to grab and throw the balls.

5 FINDINGS

All players identified the shield mechanic as a primary difference between the two games. Among the ten players, four players indicated a preference for the version of "Eye Ball" with the gaze mechanic, three players indicated no preference for either, and three players indicated a preference for the game without the gaze mechanic. We identify four themes from players' preference justification and impressions of the gaze mechanic:

- **Gaze mechanic facilitates strategy and challenge:** In the study, a significant portion of participants, particularly those expressing a preference for the version incorporating the gaze mechanic, frequently used terms like "strategy"

or "tactical" when describing their impressions of this feature. Tactical considerations encompassed both the player's own gaze behavior (using the shield to defend against the opponent's attacks) and the opponent's gaze (monitoring the opponent's gaze to better anticipate their next moves). To further emphasize this tactical element, one could enhance the auditory feedback of gaze behavior, for example, by amplifying the shield sound, to more distinctly highlight these states. Conversely, some players who did not express a strong preference for the gaze mechanic version noted an increase in complexity attributed to this mechanic.

- Gaze mechanic has a learning period:** Although players were capable of learning the new mechanic, two participants noted that it took some time "to get used to." However, this exposure to the novel mechanic was limited to a single, brief three-minute trial. To enhance the complexity and comprehensibility of the gaze mechanic, it is advisable to develop more straightforward and intuitive forms of this feature. One approach could involve aligning the concepts more closely with the meaning of the actions. For instance, making the player invisible by closing their eyes could be used to escape dangerous situations. Another method to make the gaze mechanic more understandable might include introducing the mechanic only after a certain period of gameplay and through specific events (like a Power Up). This gradual introduction would allow players to become familiar with the game mechanics step by step, and the inclusion of elements like Power Ups could offer them greater control over their abilities. Furthermore, an avenue for improving the evaluation could involve extending the gameplay duration to minimize acclimation and novelty effects.
- Multi-sensory cues could improve usability:** Several players suggested that incorporating audio cues for the motion of the ball or for the shield could enhance usability. However, one participant noted that the mechanic felt responsive with the current audio cue. This feedback underscores the importance for designers to utilize audio or haptic cues to relay information to players, especially when their vision is restricted. The activated shield was accompanied by a sound, but this was too subtle and not loud enough in comparison to the music (a humming noise). This aspect is particularly crucial for the player, rather than the opponent, as the player relies on auditory and haptic feedback when their eyes are closed. In addition to a sound, additional animation effects on the opponent's shield could inform the player that the opponent is now protected.
- Shield footprint could change to improve usability:** Several players indicated that the shield could be more effective if it had a larger blocking area. The shield was originally designed to protect only the front side of the player, allowing the opponent to score points by bouncing the ball off the edges. Enlarging the shield could make its use more attractive (e.g., a larger surface area, a spherical shape to protect the player from all sides). It's also worth considering enhancing the ball's bounce effect off the shield, enabling the player to swiftly return the ball to the opponent, potentially scoring a point. One player also observed that the version without

the shield encouraged more movement. This observation aligns well with the dilemma inherent in the original design. The player gains protection but loses visibility and mobility. This reinforces the earlier point about the need to improve the shield or make it more appealing to the player.

In response to the last question, players proposed ideas for applying the gaze mechanic in other game genres, such as first-person shooters, puzzles, and real-time strategy games. Suggestions included using gaze to influence the visibility or movement of in-game objects and to communicate with Non-Player Characters or other players. The aspect of object visibility opens up a multitude of possibilities for game design.

6 CONCLUSION AND FUTURE WORK

We set out to explore the possibility of using gaze as a social game mechanic, by creating a game with and without this mechanic and testing it out with players. Results demonstrated the potential for using gaze to establish strategic dilemmas, with participants expressing both enjoyment of the current mechanic, as well as ideas for future applications of social gaze mechanics in social VR play experiences.

Looking ahead, the integration of diverse gaze mechanics offers potential for further exploration. The concept of rendering players invisible when not directly looked at is just one of the many intriguing possibilities that could enhance VR gameplay. Moreover, there is a compelling opportunity to enhance the cues of gaze states, both visually and auditorily, to create a more immersive and intuitive gaming experience. Another promising direction is the investigation of asymmetric gameplay scenarios. Understanding how the dynamics of the game change when one player possesses the gaze skill and the other does not could yield valuable insights into the strategic depth and competitive balance of VR games. Furthermore, expanding our play tests to larger groups and different player configurations, such as 3 vs. 3 scenarios, could provide a broader understanding of social interactions and team dynamics in gaze-based VR gaming. Additionally, extending the duration of game play phases would be crucial in mitigating the novelty effect, allowing players to fully master the gaze mechanics and appreciate their strategic value over time. Finally, the application of gaze interaction concepts to different game genres could open up new horizons for game design and player engagement. This could lead to a range of novel gaming experiences that leverage the capabilities of VR technology.

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